Aligned latent-preference aggregation: a feasibility check

Handoff to Yu He · Paper B · 2026-10-13 · drafted by Fan Yao with Claude; decisions by Fan. A plan, not results: the only numbers below are background diagnostics from earlier notes. Sources: the handoff (`HANDOFF_TO_YU_HE_2026-10-13.md`), the plan `draft-v2.md` with Fan's decisions, the latent-preference and partial-sharing notes, the 2026-09-25 Hodge-diagnostics report and the 10-13 meeting minutes (Section 9).

> **In short.** Yu He fits one skew-symmetric latent preference $\bar{A}$ on response features from the pooled pairwise labels of the 500 fixed HelpSteer panels (two links, oracle 1, oracle 2 and their mixture, embedding features first), and reads the shared cycle off as the principal rotation plane of S_{eff}, with rotation rate $\tilde{\omega}_1$. A common cyclic signal counts as measurable only if, at the initial checkpoint's policy and on the held-out half of a cross-fit, $\tilde{\omega}_1$ clears the 95th percentiles of a within-prompt shuffle null and an orientation-flip null and the value a transitive surrogate produces, and the implied frontier β^* lies inside the β values the run actually used; the saved checkpoints are then projected onto that plane to tell rotation, polarization and a stationary state apart. Nothing has been run yet: the 09-25 one-hot version of the same test found aligned $\tilde{\omega}$ of 0.014–0.097, far above its shuffle null but explained by label discretization, and the oracle construction, mixing rule and run parameters still have to be confirmed from code.

Colours: latent $\bar{A}$ / shared cycle · prompts and panels · policy / checkpoints · nulls and controls · oracle 1 · oracle 2 · mixture · verdict pass · fail

1 Scope and starting point

This handoff covers the P0 item of the 10-13 meeting, plus the checkpoint projection that turns it into the per-checkpoint metric of Idea A. Everything is a CPU-only diagnostic on data and checkpoints Yu He already has: no new GPU training, no change to the training protocol, no publication.

> **Meeting minutes 2026-10-13, §7 todo, P0, owner Yu He (Fan brainstorming with him)**
> *"定义共同latent坐标/投影和合法的聚合方法，快速计算聚合cyclic magnitude；先看是否存在可测的共同信号"* — boundary in the same row: *"明确要求先做feasibility check；算法/metric尚未固定，不是大规模训练审批"*

Why now

The meeting's headline was negative: *"尚没有证明"理论中的 cycle 导致当前 neural 波动""* (minutes §1). As reported by Yu He in the meeting and not yet checked against code:

- The current run uses HelpSteer prompts with four responses each as text support, but **not** HelpSteer's attribute ratings; labels come from two newly designed oracle 1 / oracle 2, mixed (equal weights and 0.6/0.4 were tried or discussed). Among the 500 prompts only one has an explicit order cycle.
- Printed Hodge residuals $\|C\|_2$ can be large on panels without an explicit cycle, and the panels with the largest C do not show visibly stronger oscillation in the curves. No reliable full table could be recovered from the transcript, so this is not a correlation test.
- Checkpoints are saved at every step; the per-step win rate has not been computed this round.
- Oracle identities, the mixing rule (probability or hard label), full parameters and run IDs were not fixed in the meeting and must be taken from code and configs.

Fan proposed two explanations to be told apart: the per-prompt fluctuations are mostly training or evaluation noise, not driven by C; or shared parameters couple prompts, per-prompt cycle directions cancel in the shared update, and only the aligned common cyclic component matters. This plan tests the second explanation's precondition: whether an aligned common cycle exists in the current labels at all.

What already exists

The 09-25 diagnostic (`hodge_diagnostics/reports/2026-09-25/REPORT.md` , section "Cross-prompt coherence") and `hodge/coherence.py` already implement the one-hot special case of this plan, with model identity (UltraFeedback, MT-Bench) or within-prompt length rank (HelpSteer) as the shared feature. That tool is the one-hot case of Theorem L3 and extends directly to continuous features; its numbers are background for Section 4, not results of this plan.

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2 The objects

The plan needs three different objects, and the meeting asked that they not be confused.

> **Meeting minutes 2026-10-13, §2 (Idea A, the operational metric)**
> *"后续应明确：数据/oracle的固定偏好结构、每轮模型生成样本的估计矩阵、模型自身的风格分布分别是什么，不能把它们当作同一个东西"*

Object	What it is	Changes with t?	Used in
1 · Latent preference $\bar{A}$	A fixed antisymmetric matrix $\bar{A} \in \mathrm{Skew}_m$ on feature space; its attribute block $\bar{A}_{\mathrm{cyc}}$ is the **shared cycle**	No (set by the oracles)	S2, S3, S5
2 · Per-round panel matrices	P_{ij} estimated on the responses the checkpoint generated	Yes	Not used in this handoff, except S0's fixed panels
3 · Model latent state $\theta^{(t)}$	The checkpoint's logits regressed onto the shared features: the Idea A metric	Yes	S4

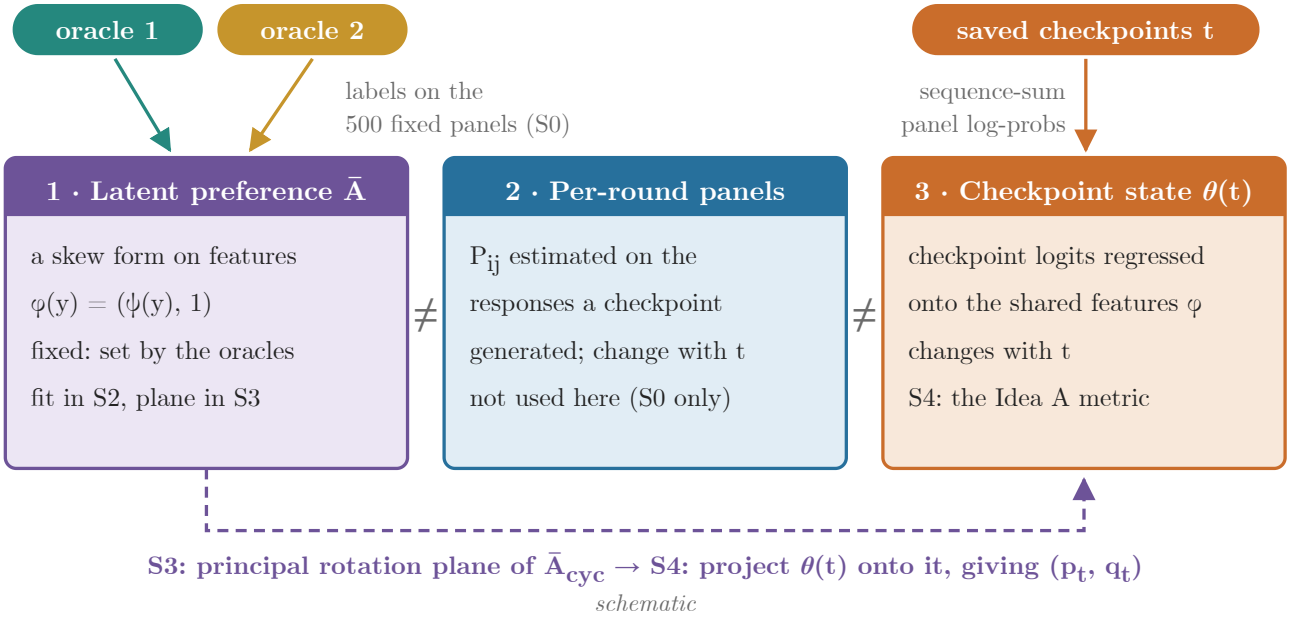


Figure 1. The plan works with three different objects: the fixed latent preference $\bar{A}$ estimated from the oracles' labels, the per-round panel matrices (left aside here), and the checkpoint state $\theta^{(t)}$, which meets $\bar{A}$ only through the S3 plane.

Definition of the latent preference

Each response y has features $\varphi(y) = (\psi(y), 1) \in \mathbb{R}^m$, where $\psi \in \mathbb{R}^{m-1}$ are prompt-independent attribute coordinates (concise, detailed, confident, ..., or a low-dimensional projection of an embedding) and the last coordinate is an intercept. The latent preference is a skew-symmetric $\bar{A}$, and the flow between any two responses, from whichever prompt, is

$$A(y, y') \;=\; \varphi(y)^\top \bar{A}\, \varphi(y').$$

- **Latent Hodge split** (Prop L1). The intercept row of $\bar{A}$ is the latent potential $\bar{u}_{\text{lat}}$, a linear score; the attribute block is the latent cycle $\bar{A}_{\text{cyc}}$.
- **Prompt-specific part.** In general $A_a = \Phi_a \bar{A} \Phi_a^\top + E_a$, where E_a is the residual that shared features cannot explain (correctness, context). The meeting's "global + specific" is this split, and the addition happens on the flow scale $\Psi(P) - \Psi(\frac{1}{2})$, not on probabilities (draft-v2 §1). This round fits the pure latent part only.
- **Gauge.** Shifting the attribute origin $\psi \to \psi + t$ maps $(\bar{u}_{\text{lat}}, \bar{A}_{\text{cyc}})$ to $(\bar{u}_{\text{lat}} - \bar{A}_{\text{cyc}} t,\ \bar{A}_{\text{cyc}})$ and leaves every panel flow unchanged. Panel centroids and latent potentials are therefore the same object; only $\bar{A}_{\text{cyc}}$ is gauge-invariant, and it is the object we want.

Each prompt sees a compression

Prompt a has a feature matrix Φ_a (one row per response), centred features $\tilde{\Phi}_a$ and attribute centroid $\bar{\psi}_a$. Its panel flow and Hodge parts are (Prop L1):

$$A_a = \Phi_a \bar{A} \Phi_a^\top, \qquad C_a = \tilde{\Phi}_a \bar{A}_{\text{cyc}} \tilde{\Phi}_a^\top, \qquad u_a = \tilde{\Phi}_a \big(\bar{u}_{\text{lat}} + \bar{A}_{\text{cyc}} \bar{\psi}_a\big).$$

So the cyclic residual C_a depends on $\bar{A}$ only through $\bar{A}_{\text{cyc}}$, and a two-response panel ($K_a = 2$) has $C_a = 0$ whatever $\bar{A}$ is. Random panels in a three-dimensional attribute space expose on average 0.000, 0.304, 0.509 and 0.667 of a pure latent cycle's energy for $K = 2, 3, 4, 6$ (latent-preference note §5(2), 200 panels each); the shares fall when the attributes have a nonzero mean.

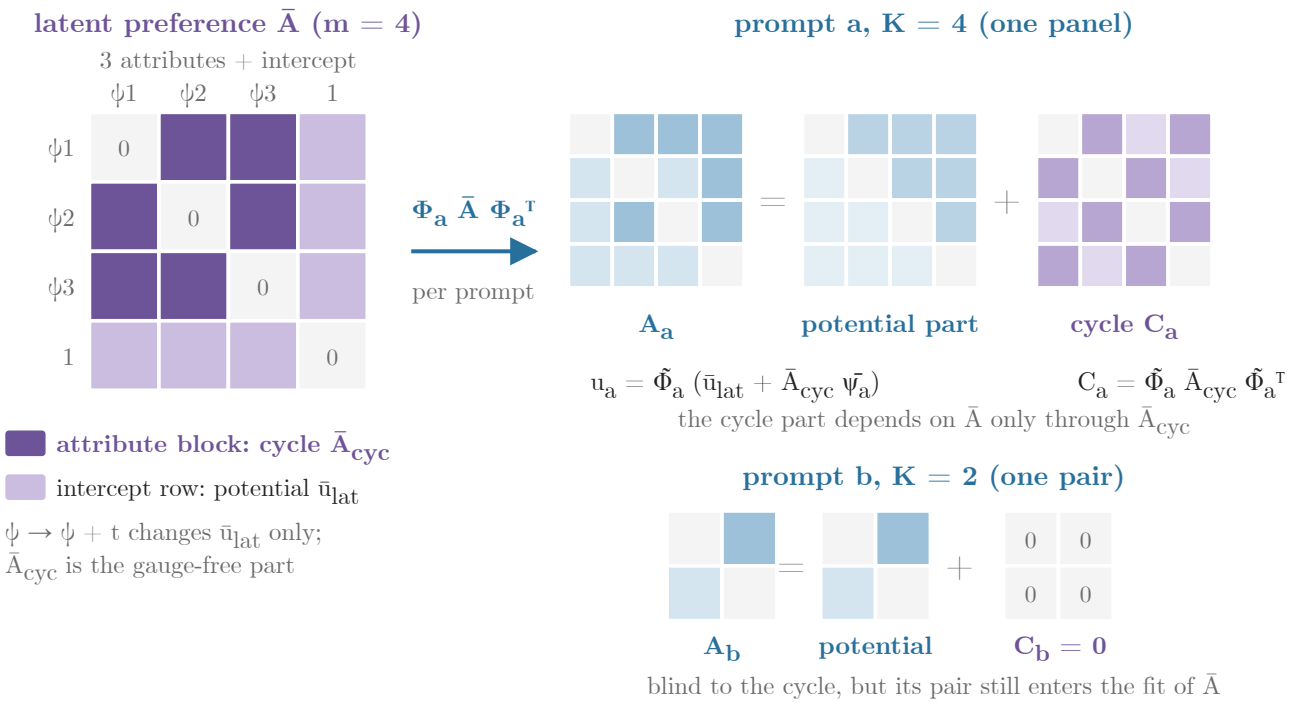


Figure 2. Each prompt sees a compression of one latent matrix: the intercept row of $\bar{A}$ becomes the panel's potential, the attribute block its cycle C_a, and a two-response panel exposes none of the cycle while still informing the fit (schematic; the bottom strip quotes the latent-preference note's simulated shares).

The legal aggregation. Response indices mean nothing across prompts, so raw response-index matrices cannot be averaged. Pulling every panel's flow back through its own Φ_a to Skew_m is the legal aggregation: it is exactly the least-squares normal equation of the latent fit in Section 3, not a hand-made alignment.

> **Meeting minutes 2026-10-13, §2, Fan's proposal, step 2**
> *"把 prompt-level preference matrices 放到可比较的共同坐标中，再聚合；不能将任意 response-index矩阵直接平均。"*

3 Estimating $\bar{A}$

The estimator is a skew-bilinear Bradley–Terry model, a standard generalized linear model in the $m(m-1)/2$ free entries of $\bar{A}$ (Prop L2). Data are pooled comparisons (a, i, j, label) from any prompt, with hard or soft labels and any sparsity.

$$P(i \succ j \mid a) = \sigma\big(\langle \bar{A}, M_{ij} \rangle\big), \qquad M_{ij} = \tfrac{1}{2}\big(\varphi_i \varphi_j^\top - \varphi_j \varphi_i^\top\big).$$

- **Logit link (DPO):** logistic regression with design row $\mathrm{vec}(M_{ij})$ and the label as target. **Identity link (IPO):** least squares on the flows. A small ridge (1e-3) is used only for conditioning.
- **Identification.** $\bar{A}$ is identified exactly when the M_{ij} of the pooled pairs span Skew_m; otherwise only its restriction to that span is. With noisy labels, consistency needs the population condition $\mathbb{E}[\mathrm{vec}M\,\mathrm{vec}M^\top] \succ 0$ on Skew_m, not merely a spanning sample.

- **Features must be continuous.** With one-hot role labels plus an intercept, each observation equals $\bar{A}_{rs} + \bar{u}_r - \bar{u}_s$, so cycle and potential are confounded (Prop L2(a), as corrected by the referee). Features must vary continuously across prompts on the same attribute. With dense features, 60 prompts with one pair each recover a 4×4 $\bar{A}$ with maximum entry error $2.0 \cdot 10^{-16}$ (design rank 6 of 6); the role-plus-intercept toy reaches design rank 3 of 6 (latent-preference note §5(1)).
- **Centring.** Attributes are centred on the pooled mean before the fit. This changes only $\bar{u}_{\text{lat}}$, not $\bar{A}_{\text{cyc}}$, and avoids gauge artefacts.

Feature set (S1)	Attribute coordinates	m	Unknowns m(m−1)/2
(b) embedding PCA, d = 8	8	9	36
(b) embedding PCA, d = 16	16	17	136
(a) judge attributes (6–8) + length	7–9	8–10	28–45

Derived from the handoff's feature choices with m = attribute coordinates + 1. Data per oracle: 500 panels × 6 pairs = 3,000 labelled pairs (derived).

The prompt-specific residual and ρ (second round)

The full model is $\sigma(\varphi_i^\top \bar{A} \varphi_j + \varepsilon_{a,ij})$ with ε skew and one block per prompt, penalized by a group lasso or ridge whose weight is the partial-sharing ρ; a ρ-scan gives the share of flow energy the shared part explains, and $\rho \to \infty$ is the pure latent model. In the partial-sharing note, ρ is the cost of prompt-specific policy components: $x_a = s + t_a$ with penalty $\frac{\rho}{2} \sum_a w_a \|t_a\|^2$, so $\rho = 0$ is the isolated model and $\rho \to \infty$ the pooled common-role model (X1), with state-dependent coupling $(J_a + \rho \Pi)^{-1} J_a$. Its latent version uses features $[\varphi_{\text{attr}};\, 1_{\text{group}}(a) \otimes \varphi_{\text{attr}}]$ with a ridge on the second block. **Fan's decision: this round fits only the pure latent model ($\rho \to \infty$); the ρ-scan is the second round.**

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4 The shared cyclic direction and its nulls

The shared cycle is not a single number but a plane: the principal rotation plane of the effective cyclic operator of Theorem L3, weighted by where each prompt's policy still has mass. At a policy π_a on each panel:

$$J_a = \operatorname{diag}(\pi_a) - \pi_a \pi_a^\top, \qquad G_a = \tilde{\Phi}_a^\top J_a \tilde{\Phi}_a, \qquad G = \sum_a w_a G_a,$$

$$S = \sum_a w_a\, G_a \bar{A}_{\text{cyc}} G_a, \qquad S_{\text{eff}} = G^{-1/2} S G^{-1/2}.$$

S_{eff} is real and skew, with eigenvalues $\pm i\tilde{\omega}_j$. The largest, $\tilde{\omega}_1 = \max |\operatorname{Im} \operatorname{eig}(S_{\text{eff}})|$, and its two-dimensional invariant plane are the most coherent cyclic direction. On the realizable branch the multipliers are $\alpha \pm i\beta\lambda\tilde{\omega}_j$, so the latent frontier is $\alpha^2 + (\beta\lambda\tilde{\omega}_1)^2 < 1$, that is

$$\beta^* = \frac{\sqrt{1 - \alpha^2}}{\lambda\, \tilde{\omega}_1}.$$

When all prompts share one Fisher matrix, $S = G \bar{A}_{\text{cyc}} G$ and the spectrum is that of the master recursion with (C, J^*) replaced by $(\bar{A}_{\text{cyc}}, G)$. For scale: the latent-preference note's toy (m = 4, $\bar{A}_{\text{cyc}} = 0.1\, R_{\text{cyc}}$, $\alpha = 0.9$, $\lambda = 1$) has S_{eff} eigenvalues ± 0.0797i and 0, hence $\beta^* = 5.47$; its shared policy is bounded and nonconvergent for $8 \le \beta \le 16$ and concentrates at $\beta = 24$.

1 **This is the corrected version of Idea B's "PCA" analogy.**

PCA finds principal directions of a symmetric matrix; here the object is the principal rotation plane of a skew matrix. Through J_a it weights each prompt automatically, so an attribute counts only if it varies on many prompts and the policy still has mass there (draft-v2 §3). The minutes called the PCA idea *"直觉类比，不是已给出的 PCA算法或正确 estimator"* (§3).

2 **A latent cycle is necessary, not sufficient, for shared instability.**

Each panel adds its induced potential $\tilde{\Phi}_a(\bar{u}_{\text{lat}} + \bar{A}_{\text{cyc}}\bar{\psi}_a)$ to the pooled forcing; unless the aggregate cancels, the shared policy concentrates and the cycle is damped, and even with zero induced potential a strong enough gain ends in concentration (latent-preference note §0 item 5, Thm L3(iv)). This is why S4 looks at the checkpoints' actual motion and not only at $\tilde{\omega}_1$.

Where to evaluate and what to report (S3)

- Two policies: the initial checkpoint's actual panel probabilities π_a (sequence-sum, as logged), which is the **primary** policy for the decision rule, and **uniform** on each panel, reported as secondary.
- Report $\tilde{\omega}_1$, its plane (two vectors in attribute space; print loadings on the interpretable attributes once set (a) exists), and β^* for the run's α and λ. Compare with the run's effective β: DPO $\beta_{\text{train}} = 0.1 \Rightarrow$ paper $\beta = 10$; IPO $\beta = 1/\tau$.
- Interpretable reading: if the loadings concentrate on two or three attributes, the result can be stated in Idea A's language (draft-v2 §3, item 3).

Nulls and validation (mandatory)

> **Meeting minutes 2026-10-13, §3, open implementation points**
> *"如何避免在同一评估数据上“最大化cycle”而选择到噪声"*

Check	What it keeps	What it destroys	Report
Within-prompt label shuffle	Every $\lvert C_a\rvert$, every label set, and G	Alignment across prompts	500 draws on the held-out half; 95th pct. of $\tilde{\omega}_1$
Random orientation flip	Each panel up to sign: $A_a \to \pm A_a$ (X4's random-direction model, expected $\bar{A}_{\text{cyc}} = 0$)	A consistent orientation	500 draws on the held-out half; 95th pct. of $\tilde{\omega}_1$
Transitive surrogate	Each panel's order: a single Bradley–Terry score per panel, pushed through the same link, then $\bar{A}$ refit	Any genuine cycle	The $\tilde{\omega}_1$ a purely transitive order produces by discretization or link alone (the 09-25 control)
Cross-fitting	Split prompts into halves; fit $\bar{A}_{\text{cyc}}$ and the plane on one half	In-sample selection of the plane	Held-out $\tilde{\omega}$ and cyclic share, prompt-level bootstrap CI

Why the surrogate is needed. A transitive oracle whose score is nonlinear in the features can leak into a fitted bilinear $\bar{A}_{\text{cyc}}$; shuffling labels within prompts and flipping orientations do not remove such a leak, and

cross-fitting does not either. The transitive surrogate measures exactly that artefact, as it did on 09-25, where it reproduced and exceeded the aligned signal on every dataset.

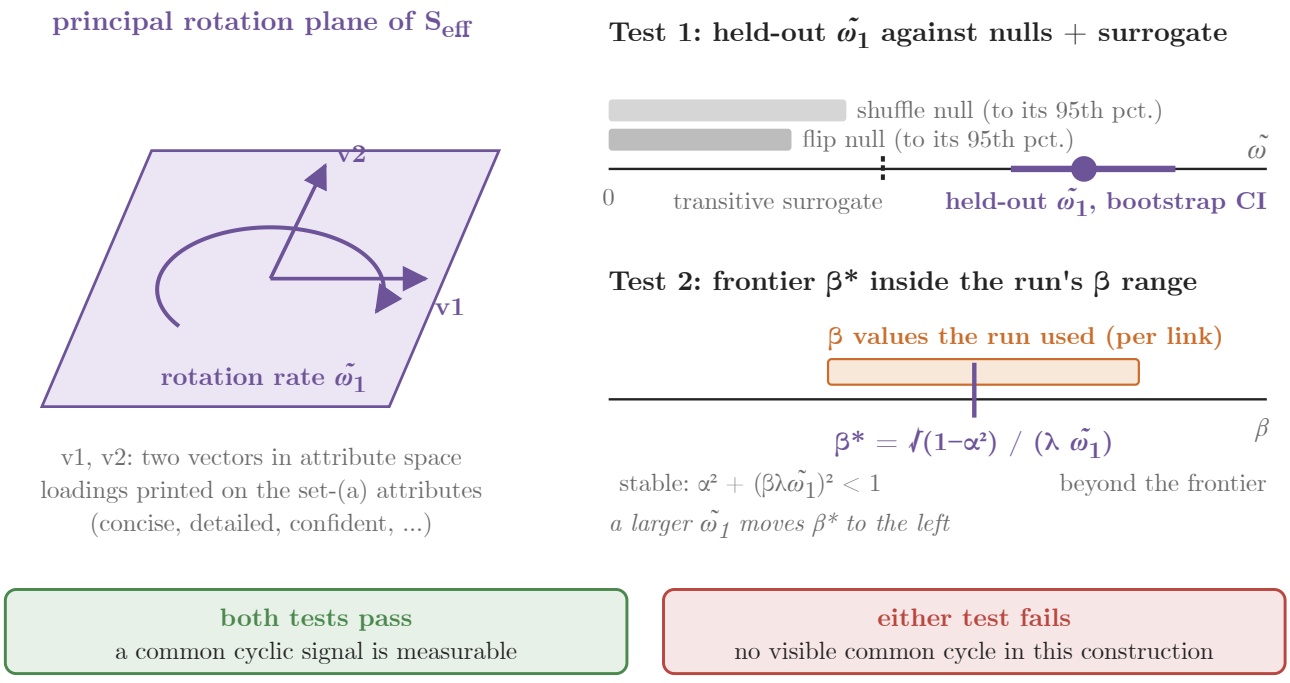


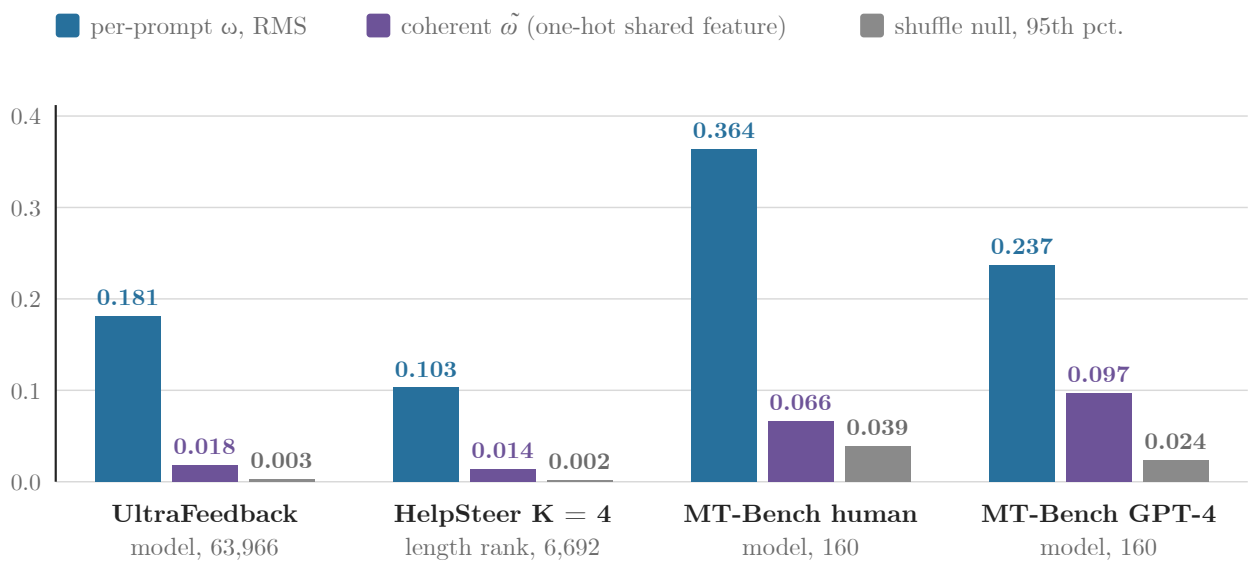
Figure 3. The shared cycle is a plane with a rotation rate $\tilde{\omega}_1$; it counts only if the held-out rate, at the initial checkpoint's policy, clears both null bands and the transitive surrogate, and the frontier it implies falls inside the β values the run used (schematic).

Background: the one-hot version on 09-25

The 09-25 report computed the same quantities with one-hot shared features (model identity or length rank), at uniform policies, under the logit link with 0.5 pseudo-votes; its $\tilde{\omega} = \max|\mathrm{Im}\,\mathrm{eig}(G^{+}S)|$ has the same spectrum as S_{eff} when G is invertible. Quoted exactly:

Panels (shared label)	per-prompt ω, RMS	coherent $\tilde{\omega}$	ratio	shuffle null, 95th pct.	transitive surrogate $\tilde{\omega}$	soft BT $\tilde{\omega}$
UltraFeedback 63,966 panels; model	0.181	0.018	0.10	0.003	0.053	0.0016
HelpSteer K = 4 6,692 panels; length rank	**0.103**	**0.014**	**0.14**	**0.002**	**0.063**	**0.0015**
MT-Bench human 160 panels; model	0.364	0.066	0.18	0.039	0.104	–
MT-Bench GPT-4 160 panels; model	0.237	0.097	0.41	0.024	0.119	–

From `hodge_diagnostics/reports/2026-09-25/REPORT.md`, "Cross-prompt coherence under parameter sharing"; not recomputed. HelpSteer is highlighted because the current run uses HelpSteer text.



uniform policies, logit link, 0.5 pseudo-votes; background from the 2026-09-25 report, not this plan

Figure 4. Background, not a result of this plan: in the 09-25 one-hot version the coherent $\tilde{\omega}$ clears its shuffle null on all four datasets but is only a tenth to two fifths of the per-prompt magnitude (numbers quoted from the 09-25 report).

- **Alignment was real but small.** Every coherent $\tilde{\omega}$ is far above its shuffle null ($p = 0.002$ on every dataset), yet the aligned part is only 10–41% of the per-prompt magnitude.
- **And it came from discretization.** A perfectly transitive surrogate with saturated labels reproduces and exceeds it on every dataset (0.053 against 0.018 on UltraFeedback); soft Bradley–Terry labels shrink it tenfold. The exact dynamics of the maximally shared policy converged in every configuration tested, with real and shuffled labels.
- **What it did not test.** Only identity features were used; coherence in style or semantic features was not excluded (09-25 caveats). That gap is what the continuous features of this plan fill. The 09-25 one-hot numbers are not to be reported as results of this plan (draft-v2 §7).

5 Steps S0–S5

All six steps run on CPU, in this order; S4 and S5 depend on S2–S3 but not on the verdict.

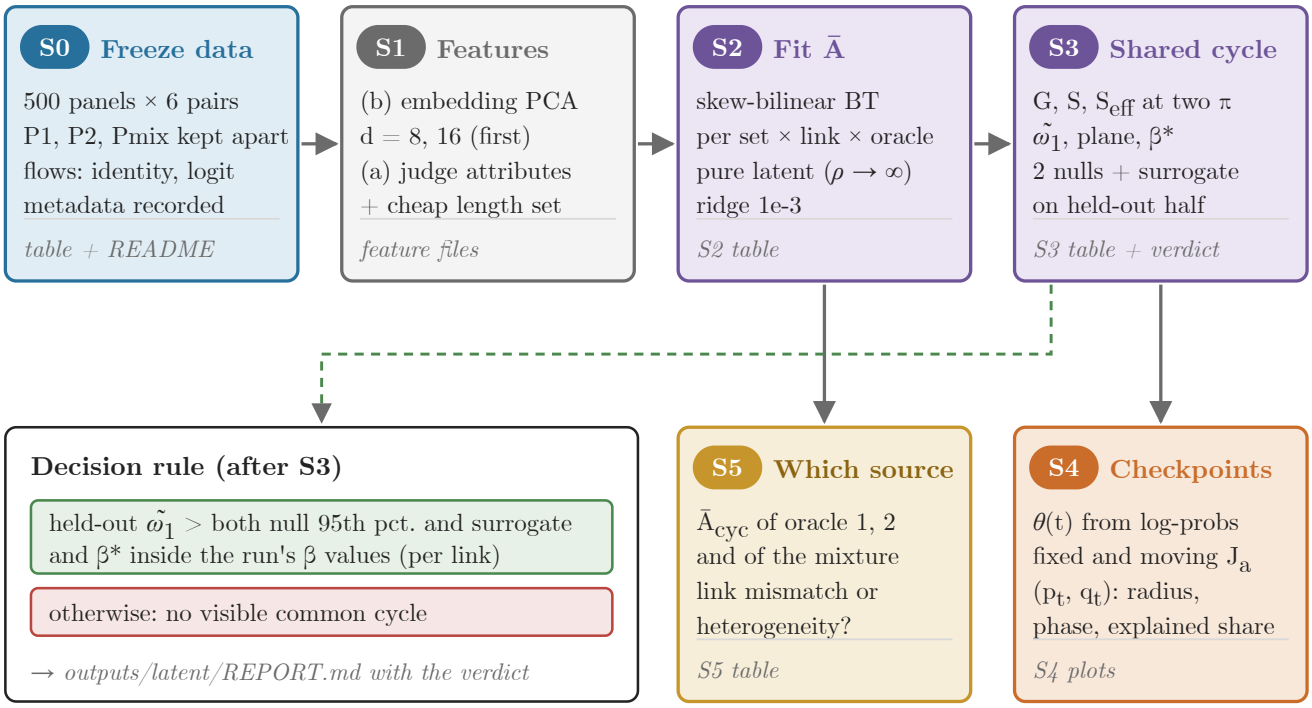


Figure 5. Six CPU steps lead from a frozen label table to one verdict: S0–S3 decide whether a common cycle is measurable, S4 asks whether the checkpoints move along it, and S5 asks whether it comes from one oracle or from the mixture.

S0 Freeze the data table

Export from the current 500-panel run, for all 6 pairs of each 4-response panel, with the two oracles' probabilities **separately** and the mixed one:

Column	Content
`prompt_id`	The prompt (panel) identifier
`i` , `j`	The two responses of the pair, within the panel
`P1_ij`	Oracle 1's probability that $i \succ j$
`P2_ij`	Oracle 2's probability that $i \succ j$
`Pmix_ij`	The mixed label actually used in training
`mix_weights`	The mixing weights
`response_text_i` , `response_text_j`	The response texts (needed for S1)

- **Record:** oracle identities; mixing rule (probability vs hard label); run and config IDs; which checkpoint generated the responses (per the meeting the responses are fixed HelpSteer text; confirm).
- **Flows under both links:** identity $A = P - \frac{1}{2}$ (IPO) and logit $A = \log \frac{P}{1-P}$ with 0.5 pseudo-counts (DPO).

- **Why the mixing rule matters:** in the 09-25 report, hard labels on a mean score read through the IPO identity link gave cyclic share 0.10 on every strictly ordered four-response panel, coming from the 0/1 labels rather than the preferences, and the same labels make DPO logit targets infinite.

Deliverable: one parquet or CSV file, plus a README line per column.

S1 Features: two sets, embedding first

- **(b) Embedding, first.** A sentence embedding of each response from any frozen encoder already on the server (record the model), pooled PCA to d $\in$ {8, 16}, centred on the pooled mean. Zero annotation cost.
- **(a) Interpretable, second.** One judge call per response (2000 responses), scoring 0–10 on about 6–8 style attributes: concise, detailed, confident/hedged, polite, structured (lists, headers), formal, example-giving; plus length in tokens. Record the judge model and the prompt verbatim; centre on the pooled mean. (Draft-v2 suggested a Nemotron or GPT-class judge and 5–8 attributes; the handoff's list is the current one.)
- **(c) Cheap set, kept for comparison.** Length, length rank, sentence count, so S2 can be set against the 09-25 one-hot result.

Feature vectors must vary continuously across prompts on the same attribute: pure one-hot role labels plus an intercept confound cycle and potential (Prop L2(a), corrected).

Deliverable: per-response feature vectors for each set, with the encoder or judge and prompt recorded.

S2 Fit $\bar{\mathbf{A}}$ (pure latent, no prompt-specific residual this round)

For each feature set × link × {oracle 1, oracle 2, mixed}: logistic regression (logit link) or least squares on flows (identity link), design row $\mathrm{vec}(\mathrm{skew}(\varphi_i\varphi_j^\top))$, target the label or flow, parameters Skew_m ($m(m-1)/2$ unknowns), ridge 1e-3 only for conditioning.

- $\bar{A}_{\mathrm{cyc}}$;
- latent cyclic share $\|\bar{A}_{\mathrm{cyc}}\|_F^2/\|\bar{A}\|_F^2$;
- design rank against $m(m-1)/2$;
- per-prompt exposed share $\|C_a\|_F^2/\|A_a\|_F^2$ with $C_a = \tilde{\Phi}_a \bar{A}_{\mathrm{cyc}} \tilde{\Phi}_a^\top$.

The ρ-scan (shared plus prompt-specific residual with ridge ρ, the partial-sharing construction) is the second round, not now.

Deliverable: S2 table, one row per feature set × link × oracle.

S3 The shared cyclic direction and its null

At the uniform policy and at the initial checkpoint's π_a, compute G, S, S_{eff} (Section 4); report $\tilde{\omega}_1$, its plane and loadings, β^* and the run's effective β; run the cross-fit, and evaluate $\tilde{\omega}_1$, the shuffle and flip nulls (500 draws each, 95th percentile) and the transitive surrogate on the held-out half, with a prompt-level bootstrap CI. List the β values the run actually used, per link.

Deliverable: S3 table with $\tilde{\omega}_1$, nulls, transitive surrogate, held-out CI, β^*, loadings; the verdict of Section 6.

S4 Checkpoint projection (the Idea A metric)

For every saved checkpoint t and the same 500 panels, take the sequence-sum log-probabilities of the 4 fixed responses, centre them within the panel ($\tilde{x}_a^{(t)}$), and solve

$$\theta^{(t)} = \arg\min_{\theta} \sum_a w_a \left\| \tilde{\Phi}_a \theta - \tilde{x}_a^{(t)} \right\|_{J_a}^2 .$$

- **Both metrics, as Fan decided:** J_a from the initial checkpoint (fixed metric) and J_a from checkpoint t (moving metric).
- Project $\theta^{(t)}$ onto the S3 plane $\to (p_t, q_t)$; plot radius and phase over t, and the explained share $\sum_a \|\tilde{\Phi}_a\theta\|_{J_a}^2 / \sum_a \|\tilde{x}_a\|_{J_a}^2$. Draft-v2 also asks for the residual $\|\tilde{x}_a - \tilde{\Phi}_a\theta\|$ as a share: how much of the policy's motion the latent coordinates explain.
- No re-scoring: only panel log-probs, which the pipeline already logs (PROTOCOLS.md: sequence-sum panel probabilities and panel mass per round).
- Run for both links' $\bar{A}$ and both feature sets.

Deliverable: S4 plots (radius and phase against t, explained share) for each variant.

S5 Where the cycle comes from

Compare $\bar{A}_{\text{cyc}}$ from oracle 1, oracle 2 and the mixture, under both links. A single Bradley–Terry oracle has zero residual under the logit link and a positive one under the identity link (09-25 report, summary item 1); probability mixing is not logit-additive. Also run a transitive-surrogate control: fit a single BT score per panel, push it through the same link, refit $\bar{A}$; any $\bar{A}_{\text{cyc}}$ it produces is a discretization or link artefact, not a shared cycle. This step decides which source the paper should name (decision table in Section 6).

Deliverable: S5 comparison table.

Deliverables and code

Item	Content
`outputs/latent/`	S0 table; S2 tables (one row per feature set × link × oracle); S3 table with $\tilde{\omega}_1$, nulls, held-out CI, β^*, loadings; S4 plots (radius and phase against t, explained share) for each variant; S5 comparison table
`REPORT.md`	Short; the decision-rule verdict and the exact commands and configs
`hodge/latent.py` (e.g.)	A module next to `hodge/coherence.py`; reuse `core.flow`, `core.hodge`, `core.softmax_jacobian`; a `permutation_null`-style function for both nulls
Tests	On a synthetic $\bar{A}$: recover $\bar{A}_{\text{cyc}}$ to 1e-8 from 60 prompts × 1 pair with dense features; the $K = 2 \Rightarrow C_a = 0$ identity
Untouched	The existing six GPU runs and the per-step win-rate task run in parallel; they are not replaced

6 Decision rule and outcomes

Measurable common cycle

At the initial checkpoint's policy (primary), the held-out $\tilde{\omega}_1$ exceeds **both** null 95th percentiles (500 draws each, held-out half) **and** the transitive-surrogate value, **and** β^* lies inside the β values the run used for that link (DPO: paper $\beta = 1/\beta_{\text{train}}$; IPO: $\beta = 1/\tau$).

No visible common cycle

Otherwise: the current two-oracle construction does not supply a visible common cycle (the meeting's conditional reading).

> **Meeting minutes 2026-10-13, §3, the conditional reading**
> *"聚合 C 很小时，当前数据构造可能本来就不容易显示循环；聚合 C 不小时，再看训练是否产生对应运动。"* The same paragraph adds that a missing motion could be masked by noise, but noise is not the only explanation.

What each combination would mean (draft-v2 §5)

S3	S4	Reading	Next
✓ signal	phase advances	The strongest empirical layer for Paper B Section 7	Write the latent framework into the main text
✓ signal	stationary	Per the partial-sharing theorem, model capacity (small ρ) keeps the cycles "each in its own place"	Estimate the actual ρ from cross-prompt LoRA gradient overlap; shared-LoRA vs per-group-adapter comparison
✗ no signal	–	The current two-oracle construction does not produce a common cycle	Either a latent cyclic oracle with a known $\bar{A}_{\text{cyc}}$ (not authorized; discuss only then), or accept HelpSteer-type data as the "concentration without oscillation" negative control

The latent cyclic oracle would define $\bar{A}$ in attribute space, identical for every prompt: the continuous version of the cyclic-history pilot.

S4: three distinguishable outcomes

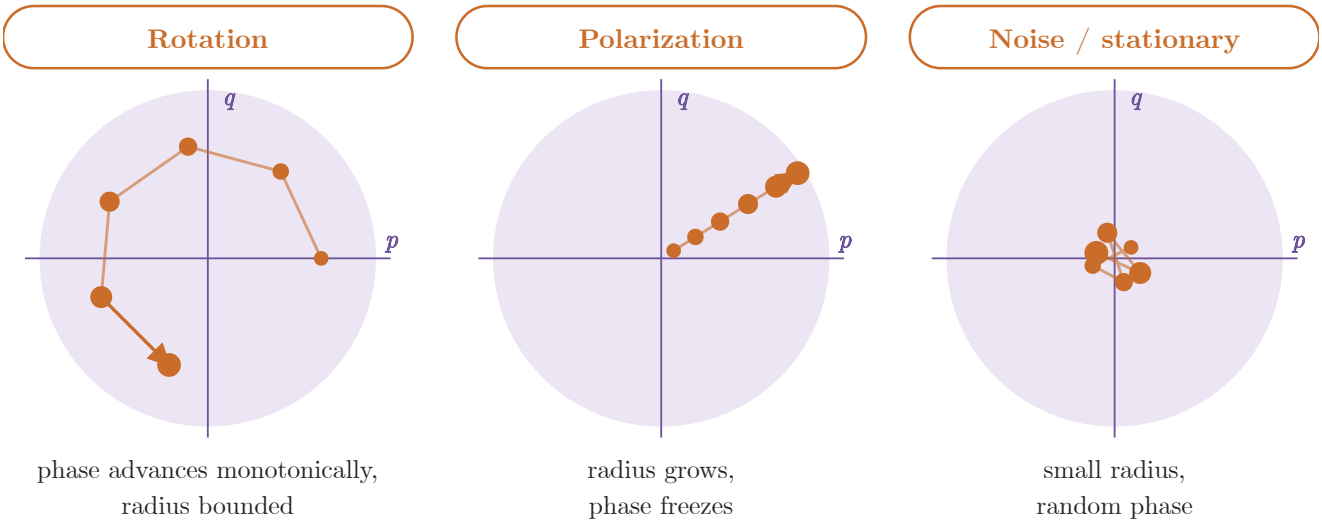


Figure 6. The checkpoint projection separates three behaviours by radius and phase over t: checkpoints circling the shared plane, running away along one direction, or jittering near the origin (schematic).

S5: which source

Observation	Source
(i) A single oracle has $\bar{A}_{\text{cyc}} \neq 0$ under the **identity** link but ≈ 0 under logit	**Link mismatch:** a transitive BT oracle read through the IPO link
(ii) A single oracle has $\bar{A}_{\text{cyc}} \neq 0$ under the **logit** link	That oracle is **not BT-transitive in the feature coordinates** (its own scoring is nonlinear or inconsistent)
(iii) Only the **mixture** has $\bar{A}_{\text{cyc}} \neq 0$ under the logit link	**Heterogeneity** between the two oracles (probability mixing is not logit-additive)
control Transitive surrogate produces $\bar{A}_{\text{cyc}}$	Discretization or link artefact, not a shared cycle

Link mismatch and heterogeneity are the two sources named in the 09-25 report; there, a single Bradley–Terry score gave an exactly zero residual under the logit link and a small positive one under the identity link.

7 Boundaries, decisions and questions

Fan's decisions (2026-10-13)

1. Features: both the interpretable attributes and the embedding; the embedding reports first.
2. S4: report both the initial-checkpoint J_a and the per-checkpoint J_a.
3. First round: pure latent fit ($\rho \to \infty$) only; the ρ-scan (shared/private split) is the second round.
4. Latent cyclic oracle (new training): not authorized; revisit only if S3 is negative.
5. Both links: identity (IPO) and logit (DPO).

What this does not authorize

> **Out of scope.** No new training runs; no latent cyclic oracle design (discussed only if S3 is negative); no ρ-scan this round; no manuscript edits; no change to the training protocol; no new GPU. The 09-25 one-hot result is not this plan's result. The effect of estimation error in $\bar{A}$ on the dynamics ($\bar{A}$ is treated as known) is left for later. Numbers from this handoff are diagnostics of data and checkpoints, not neural stability results.

The meeting also kept levels apart: an explicit order cycle, a nonzero Hodge C, a large $\|C\|_2$ and unstable outer dynamics are different things, and the spoken shorthand "identity = IPO, logit = DPO" does not remove the known gap between pairwise DPO and entrywise-logit ΨPO (minutes §1).

Questions to send back early

1. Which encoder and judge are available on the server without new access? Any constraint on calling an external judge for 2000 responses?
2. Confirm that the 500 panels use fixed HelpSteer responses (not regenerated per round) and that panel log-probs exist for every checkpoint.
3. The α, λ and β_{train} of the run whose checkpoints will be projected.

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8 Limits

- This is a plan. No quantity in it has been estimated on the current data; every figure except Figure 4 is a schematic.
- The description of the current run (oracles, mixing, one explicit cycle in 500 panels, saved checkpoints, missing per-step win rate) is a participant report from the 10-13 meeting, not checked against code or configs.
- The background numbers (Figure 4, the 09-25 table, the exposed shares 0.304 / 0.509 / 0.667, $\beta^* = 5.47$) are quoted from earlier notes and not recomputed here; the 09-25 numbers use one-hot features, uniform policies and the logit link only.
- The frontier β^* comes from the realizable, lazy linear-feature picture of Theorem L3 at a fixed point; checkpoints are not at a fixed point, LoRA features drift, and $\bar{A}$ is treated as known.
- Cross-fitting halves the 500 prompts; with up to 136 unknowns per fit (d = 16) and 1,500 pairs per half (derived), the held-out CI may be wide.
- Interpretable loadings depend on the judge and its prompt; embedding PCA directions may have no clear human meaning (minutes §3).

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9 Sources

All read on 2026-10-13. Every number and claim, with its section and status, is in `sources.md` next to this report.

File	Used for
`../HANDOFF_TO_YU_HE_2026-10-13.md` read	The whole plan: objects, S0–S5, decision rule, deliverables, boundaries, questions
`../draft-v2.md` read	Definitions and gauge (§1), estimation and ρ (§2), the three levels and the nulls (§3), the decision tree (§5), Fan's decisions (§6), exclusions (§7)
`../../latent-preference/latent_preference_framework.md` read	Props L1–L2, Thm L3, the numbers of §5 ($2.0 \cdot 10^{-16}$; 0.304 / 0.509 / 0.667; ±0.0797i, $\beta^* = 5.47$)

File	Used for
`../../partial-sharing/partial_sharing_theorem.md` §0–1 read	The ρ construction and its limits
`hodge_diagnostics/reports/2026-09-25/REPORT.md` read	The coherence table and bullets; Summary items 1, 6, 7; caveats
`meetings/2026-10-13-latent-evaluation-and-preference-shift.md` §1–3, §7 participant reports	The P0 item, the current-run description, Ideas A and B, the quotes

Derived numbers (parameter counts, 3,000 and 1,500 pairs, the 0.014–0.097 range) and the rationale sentence for the transitive surrogate in Section 4 are additions of this report, confirmed by the coordinator; everything else follows the sources.